

Disaster or Opportunity: The Impact of the Pandemic on Graduate Employment

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I. INTRODUCTION

This study examines data from the US Department of Education to see if there were changes in student salaries before and after the COVID-19 pandemic. We were particularly interested in seeing if there were any specific changes or trends emerging post-pandemic, such as certain fields adapting well and if established high-paying majors widened the earnings gap during this period. To explore this topic, we used a variety of machine learning techniques such as k-means clustering, to find specific groups that shifted in salaries during the pandemic, predictive modeling via synthetic control to compare predicted salary in a world without the pandemic, and regression analysis to examine the relationship between specific factors on salary. Our analysis found that, although there were some groups that experienced a shift in their salaries & fields during the pandemic, overall the impact of the pandemic on the job market was not significant, and resembled more of a structural overhaul, leading to changes and transformations in certain industries.

II. LITERATURE REVIEW

The economic impacts of the COVID-19 pandemic have been far-reaching, reshaping U.S. industries and altering long-term employment trends for college graduates. Some sectors, like healthcare, information technology, and engineering, proved remarkably resilient, largely due to the rapid shift to digital platforms and increased demand for healthcare services [1]. Healthcare’s critical role during the crisis ensured stability and even growth, while IT and engineering thrived as remote work and online education technologies became the new normal[4]. For graduates, these sectors now offer stronger job security and higher earning potential.

On the other hand, labor-intensive and service-oriented industries faced significant disruptions. As a general trend, unemployment surged in sectors that relied on close-contact activities. Between January and July 2020, nonfarm employment dropped by over 12.5 million jobs [4]. At the same time, consumer behavior shifted dramatically. Spending in high-contact industries plummeted by 60%, driven more by personal health concerns than official lockdowns [4].

Lockdowns hit tourism, transportation, retail, and entertainment especially hard, with small business revenue in leisure and hospitality dropping a staggering 47.5% between January and August 2020 [3]. The tourism industry, being particularly vulnerable, played a major role in the economic downturn. For every \$1 million decrease in tourism receipts, the economy lost \$1.53 million in GDP and about 17 jobs

[5]. In fact, tourism’s decline accounted for a quarter of the U.S. recession in 2020 [5].

Interestingly, while healthcare is often considered recession-proof, it experienced its own paradoxical challenges. The surge in COVID-19-related care was offset by steep declines in elective procedures and routine visits. Cutler [6] noted a 70% drop in primary care visits during the early months of the pandemic, as patients postponed non-essential healthcare out of safety concerns. Smaller healthcare practices were hit particularly hard, grappling with financial stress, furloughs, and pay cuts.

Looking ahead, the pandemic’s long-term impacts will likely include structural shifts in employment, earnings, and workforce participation—especially for new college graduates. Industries that embraced digital adaptability are better positioned to recover, offering stable job prospects and wage growth. In contrast, sectors dependent on physical interactions may continue to face challenges as they adapt to evolving consumer habits and operational constraints.

III. DATA

CONTROL	Working Sector
$EARN_M DN_1YR, 4YR, 5YR$	Median salary after 1 year, 4 years, 5 years
DISTANCE	Geographical distance
CIPDESC	Industry/Field of Work
CREDESC	Highest education attained
INSTNM	Institution where degree was earned
OPEID6	Office of Postsecondary Education Identifier (6 digits)
UNITID	Integrated Postsecondary Education Data System ID Number
CIPCODE	Four-digit Classification of Instructional Programs Code

TABLE I

KEY VARIABLES PRESENT IN DATASETS

We utilized several datasets from the US Department of Education’s College Scorecard (Link), an open source platform that provides information about institutions and fields of study, as well as post secondary costs, graduation rates, student loan debt, post-college earnings, and more. Some of the datasets used included Most-Recent-Cohorts-Fields-of-Study.csv, which looks at a variety of industries & institutions, and contains 229188 rows with 174 columns representing many different variables, alongside similar datasets from previous years. Some of the key features that we

worked with in our analysis included Control, which looks at the working sector, the median salary earned, and geographical distance, as well as degrees and institutions of education.



Fig. 1. Top Earnings by Field of Study Pre-COVID and Post-COVID

The fields of study with the highest earnings before and after the pandemic are the same, including fields such as business & nursing administration, as well as health services and computer science and information technology, as seen in Figure 1.

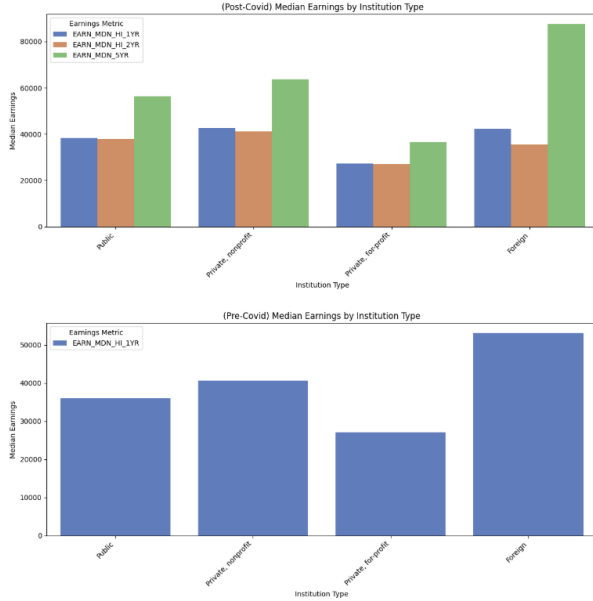


Fig. 2. Top Earnings by Institution Type Pre-COVID and Post-COVID

Similarly, as seen in Figure 2, the top earnings for institutions before & after the pandemic came from foreign institutions, followed by private non-profit, public and private for-profit.

IV. HYPOTHESIS

High-Demand Fields: Graduates in technology and healthcare fields are hypothesized to have higher starting salaries due to increased demand during and after the pandemic. Graduates with tech-focused skills (e.g., data science, software development) are expected to secure higher starting salaries, reflecting a shift toward digital skills in the job market.

Regional Influence: It is hypothesized that graduates from urban areas colleges experience higher starting salaries, benefiting from a higher concentration of employers in high-demand sectors.

Field-Specific Declines: Starting salaries in lower-demand fields, such as humanities, are hypothesized to have stagnated or declined, indicating a field-specific shift in job market demand.

Institution Type Influence: Graduates from private colleges are hypothesized to achieve higher starting salaries than those from public institutions, potentially due to differences in networking opportunities, resources, or perceived prestige.

These hypotheses guided the analysis of salary trends by fields of study, college location, and institution type, providing insights into how the job market has adapted to shifting demands.

V. METHODS

A. Clustering Analysis

K-Means clustering is an unsupervised learning technique that divides a dataset into k clusters and assigns data points to one of them depending on their distance from the center of the clusters. The primary objective of this method is to divide the dataset into a specific number of groups for comparison. We wanted to use k -means clustering to see if there were certain groups that were affected by the pandemic and may have seen a significant change in their salary. Within the dataset of Most-Recent-Cohorts-Fields-of-Study.csv we defined 4 unique clusters and grouped graduates according to features such as Distance, Control, CIPDESC, and Salary. Pre-COVID groups were marked with median salaries after 1 year, while post-COVID groups were marked with salaries after 4 years and 5 years.

B. Synthetic Control

Synthetic control analysis is a causal inference method that constructs a synthetic version of the treatment group as a weighted combination of control group observations to estimate the counterfactual outcome in the absence of treatment [7]. In recent years, advanced machine learning models such as XGBoost have been leveraged to enhance the construction of synthetic controls, providing more flexible and accurate estimations. The method involves using features from both treated and control units as input to train an XGBoost regression model, where the treated unit's pre-treatment data serves as the target. By minimizing prediction errors during the training phase, XGBoost assigns optimal weights to control units, effectively creating a synthetic counterpart that mimics the treated unit's pre-treatment trends.

The key assumption for synthetic control analysis is the Parallel Trends Assumption, which requires that the synthetic control reliably captures the untreated trajectory of the treated unit. The advantage of integrating XGBoost into synthetic control analysis lies in its ability to handle high-dimensional data, incorporate non-linear relationships, and automatically select relevant features, thereby reducing biases from omitted variables. This approach allows researchers to better estimate the causal effect of interventions, especially when traditional methods fail due to complex or noisy datasets [8].

We used this approach to investigate how the COVID-19 pandemic influenced the employment salaries of college graduates. In our study, the treatment was the onset of the pandemic, which we defined as the beginning of 2020. To estimate the counterfactual (the scenario where the pandemic did not occur), we trained an XGBoost model on data from 2010 to 2017 and used the model to predict expected salaries for the period of 2019–2023. This allowed us to compare the predicted (synthetic control) and actual outcomes to quantify the impact of the pandemic.

The primary target variable was the first-year salary of college graduates, and the predictors included factors such as undergraduate institution, academic performance, region, major, and local economic conditions. The data was sourced from the College Scorecard, which provides comprehensive annual statistics on U.S. college graduates. To ensure data quality, we excluded samples missing critical variables and derived additional predictors through calculations based on available variables.

Before conducting the analysis, we checked the validity of the Parallel Trends Assumption by comparing the pre-pandemic trends in actual salaries and model-predicted salaries. The trends aligned closely, confirming the assumption held.

Our findings revealed that the pandemic had a significant negative impact on college graduate salaries during its initial phase in 2020. However, this effect diminished by 2021, and by 2022, salaries exceeded model expectations. This suggests that while the pandemic temporarily suppressed salaries, it ultimately led to a rebound effect, with salaries surpassing predicted levels in its later stages.

C. Linear Regression

The initial step involved importing the dataset `Most-Recent-Cohorts-Field-of-Study.csv` and merging it with location-specific data from another dataset using the common `UNITID` column. This allowed us to enrich the data with information about the `CITY` and `STABBR` (state abbreviation) corresponding to each educational institution. The merged `DataFrame` (`df_merged_12`) included details such as institutional control (`CONTROL`), city, state, field of study descriptions (`CIPDESC`), and median earnings after one, four, and five years of graduation.

We extracted columns required for analysis, specifically for median earnings one year after graduation (`EARN_MDN_1YR`). The extracted data was filtered to

exclude rows where the target variable had missing or invalid values, replacing placeholders like 'PS' with `NaN` and subsequently dropping such rows. Numerical conversion ensured that all entries in `EARN_MDN_1YR` were suitable for regression analysis.

Categorical variables (`CONTROL`, `CITY`, `STABBR`, and `CIPDESC`) were transformed into numeric format using one-hot encoding. This encoding created binary columns for each category, simplifying their inclusion as predictors in the regression model. To address potential data inconsistency, all features were verified to ensure numeric format. Missing values in predictors were handled by dropping rows with `NaN`, ensuring consistency between predictor (`X`) and response (`Y`) variables.

To analyze the relationship between institutional and regional factors and graduate earnings, we fit an Ordinary Least Squares (OLS) regression model using the `statsmodels` library. A constant term was added to the predictors to account for the intercept. The model's summary provided p-values and coefficients for each variable, identifying statistically significant predictors (p-value < 0.05).

Significant predictors and their corresponding coefficients were extracted into a summary `DataFrame`, sorted by p-value for clarity. This step highlighted the most influential factors impacting graduates' earnings one year after graduation.

For comparative analysis, historical datasets from prior years were imported and concatenated into a single `DataFrame`. These datasets contained similar information regarding median earnings and institutional characteristics. The merged historical data was preprocessed and encoded using the same steps applied to the recent dataset.

An OLS model was fitted on the historical data to evaluate changes over time. The model summary and extracted significant predictors were compared to the recent data model to identify shifts in factors influencing graduate earnings.

For both current and historical datasets, the significant variables (p-value < 0.05) and their coefficients were analyzed to understand their impact on earnings. Factors with consistent significance across time were noted as stable predictors, while those showing variation highlighted changing economic or institutional dynamics. This comparison allowed for an in-depth understanding of how graduate earnings have been influenced over different periods.

The dual regression analysis provided insights into the determinants of graduate earnings across time. By leveraging both recent and historical datasets, we identified key institutional and regional predictors, highlighting shifts in their impact over time. This approach underscored the value of statistical modeling in education policy and workforce analysis.

VI. RESULTS

A. Clustering Analysis

Cluster	CONTROL	DISTANCE	CIPDESC	EARN-MDN-1YR
0	2.735620	1.076030	258.061118	0.012225
1	1.328761	0.890837	118.465215	-0.663694
2	2.179845	2.755048	147.199094	0.337699
3	2.824635	1.057498	73.839770	0.238149

TABLE II
CLUSTERS PRE-PANDEMIC

Cluster	CON- TROL	DIST- ANCE	CIP- DESC	EARN- MDN- 4YR	EARN- MDN- 5YR
0	2.738	1.252	259.535	-0.241	-0.238
1	2.446	1.297	151.243	1.778	1.768
2	2.744	1.336	76.267	-0.134	-0.126
3	1.171	1.178	122.053	-0.786	-0.799

TABLE III
CLUSTERS POST-PANDEMIC

The k-means clustering algorithm yielded 4 different groups with varying changes from pre & post COVID. Cluster 0 showed control & CIPDESC remained constant, although earnings decreased post-pandemic. In Cluster 1, all categories experienced a drastic increase from pre to post-pandemic, but in Cluster 2, Control increased but all other variables decreased. In Cluster 3, Control and salary significantly decreased, while the other two variables increased.

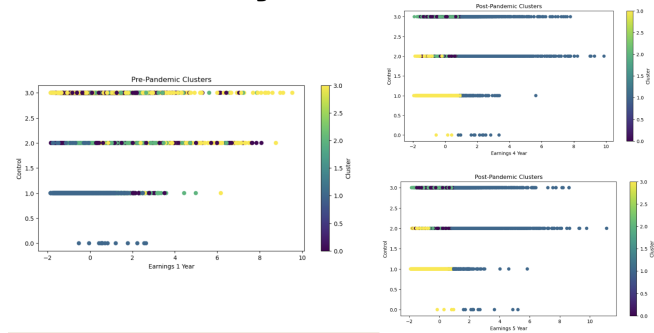


Fig. 3. Clusters - Control

This change is further shown in figures 3, 4 and 5, where there is a significant shift towards Cluster 1 from pre and post-pandemic across features such as Control, Distance & CIPDESC.

The 4 clusters seem to represent different groups that were affected financially by the pandemic. Cluster 1 experienced significant improvements, especially in earnings, and the increase in Control & Distance indicates some sort of adaptive shift during the pandemic. Cluster 0 showed a slight decrease in earnings, but not much change overall, indicating that they were not hit as hard by the pandemic but did not experience improvement either. However, Clusters 2 & 3 seemed to have been negatively impacted by the pandemic, as they had the

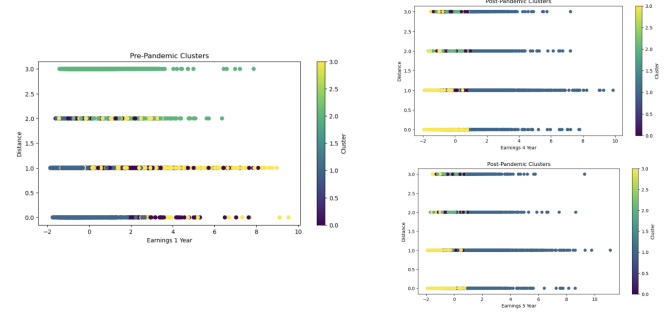


Fig. 4. Clusters - Distance

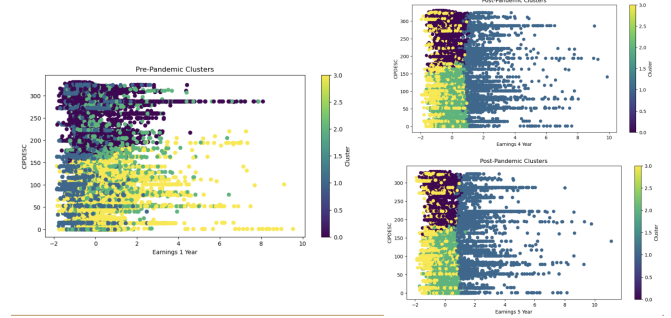


Fig. 5. Clusters - CIPDESC

biggest change in all categories & decrease in salaries. Some reasons for this could be indicated by the change in Control & CIPDESC particularly. The decrease in control for cluster 3 shows a skew towards the public sector and could thus indicate more governmental influence post-pandemic, while change in CIPDESC of both clusters 2 & 3 could indicate some sort of socio-economic shift into different industries.

B. Synthetic Control

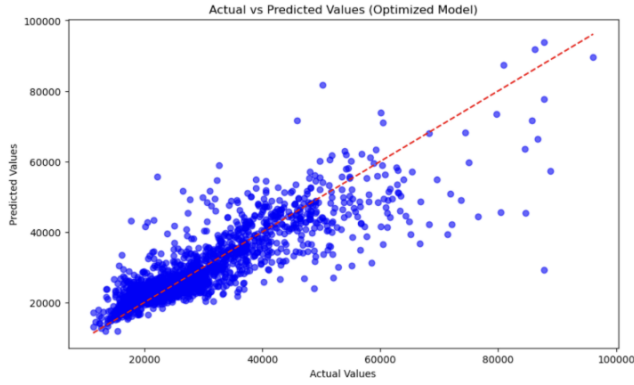


Fig. 6. Optimized Prediction Model

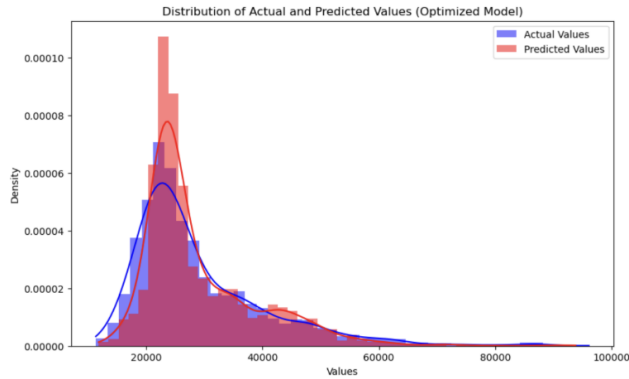


Fig. 7. Actual and Predicted Values in Optimized Model

We used a synthetic control approach powered by an XGBoost model to investigate how the COVID-19 pandemic influenced the first-year salaries of college graduates. The key input variables, such as undergraduate institution, major, academic performance, region, and economic conditions, were selected through feature engineering and feature importance testing to ensure the most optimal predictors for our model. Using these variables, we trained the model on data from 2010 to 2017 to predict average graduate salaries during 2019–2023 as if the pandemic had not occurred. The model achieved a coefficient of determination of 0.74, with its performance validated by a goodness-of-fit graph (Figure 6).

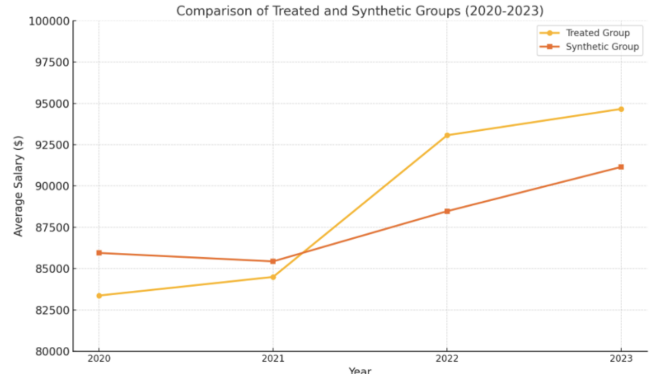


Fig. 8. Comparison of Actual vs. Predicted Salaries

Our analysis revealed that the actual salary levels in 2020 were 4.7% lower than the predicted synthetic control salaries, indicating a significant negative impact of the pandemic during its initial year. However, this divergence was short-lived; by 2021, the difference had disappeared, and in 2022 and 2023, actual salary levels exceeded the predicted levels from the synthetic control by a noticeable margin. These trends are summarized in Figure 8, which compares actual and predicted salary levels over the period.

The results suggest several potential mechanisms for these observations. The initial decline in salaries during 2020 may be attributed to widespread economic uncertainty, reduced hiring budgets, and delayed recruitment cycles caused by the pandemic's early disruptions. Companies were likely cautious in offering competitive salaries to new graduates, particularly as many industries experienced declines in revenue and faced operational challenges.

However, the subsequent recovery and surpassing of predicted salary levels in later years may reflect structural adjustments in the labor market. For example, the accelerated adoption of remote work and digital transformation during the pandemic likely increased demand for graduates in technology and other high-growth sectors. Additionally, labor shortages and heightened competition for skilled workers may have pushed employers to offer higher salaries to attract talent. These shifts suggest that while the pandemic introduced short-term salary suppression, it also catalyzed longer-term changes that benefited certain cohorts of graduates.

In summary, the pandemic's impact on college graduate salaries was complex, with a sharp but temporary decline followed by a robust recovery and eventual outperformance relative to pre-pandemic expectations. This nuanced dynamic underscores the importance of combining advanced modeling techniques like synthetic control with contextual analysis to understand causal effects in volatile environments.

C. Linear Regression

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                        OLS Regression Results
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Dep. Variable:          EARN_MDN_1YR      R-squared:                0.616
Model:                  OLS               Adj. R-squared:           0.598
Method:                 Least Squares      F-statistic:              33.56
Date:                   Thu, 05 Dec 2024   Prob (F-statistic):       0.00
Time:                   16:48:45           Log-Likelihood:           -5.8326e+05
No. Observations:       53307             AIC:                     1.171e+06
Df Residuals:           50872             BIC:                     1.193e+06
Df Model:               2434
Covariance Type:        nonrobust

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Fig. 9. Pre-COVID Model

We conducted a multiple regression analysis to explore factors influencing median earnings one year after graduation, prior to the COVID-19 pandemic. The dependent variable was median earnings after one year of graduation, with independent variables including the type, city, state of institution, and the program of study. The baseline categories were set as foreign institutions, Artesia (city), Alaska (state), and Accounting (program), allowing interpretation of each factor relative to these references. The model yielded an R-squared value of approximately 62%, indicating that the model explained 62% of the variation in median earnings, which represents a moderately good fit to the data. Furthermore, the probability of the F-statistic was very small, confirming the overall statistical significance of the model.

The analysis revealed that several programs and cities had the largest positive coefficients among all significant variables in the model. These significant positive coefficients suggest that graduates from certain cities, states, or programs earn significantly more than those associated with Accounting programs, the city of Artesia, or the state of Alaska. The top-earning programs tended to be in medical or STEM fields, while high-earning cities were often technology hubs or home to prestigious universities and medical institutions.

Conversely, the smallest significant negative coefficients indicated that certain programs and cities were associated with lower earnings compared to Accounting programs. Fields such as creative arts, liberal arts, and niche majors in relatively slow-growing industries demonstrated a salary disadvantage. The cities with negative coefficients were typically agricultural or rural areas with fewer high-paying job opportunities for graduates. Interestingly, even Cupertino, a notable tech hub, exhibited a negative association with median earnings, which might be attributable to high living costs and entry-level salary structures. Graduates from private, for-profit institutions earned, on average, more than \$10,000 less than those from foreign institutions, while graduates from public institutions earned over \$6,000 less.

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                        OLS Regression Results
=====
Dep. Variable:          EARN_MDN_1YR      R-squared:                0.619
Model:                  OLS               Adj. R-squared:           0.600
Method:                 Least Squares      F-statistic:              33.22
Date:                   Tue, 03 Dec 2024   Prob (F-statistic):       0.00
Time:                   04:03:18           Log-Likelihood:           -5.7411e+05
No. Observations:       52005             AIC:                     1.153e+06
Df Residuals:           49580             BIC:                     1.175e+06
Df Model:               2424
Covariance Type:        nonrobust

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Fig. 10. Post-COVID Model

For comparison, a similar multiple regression analysis was conducted using post-COVID-19 data. The structure of the model remained identical, with an R-squared value also around 62%, suggesting that the model's explanatory power did not diminish. The probability of the F-statistic remained very small, reaffirming the model's statistical significance. The largest significant positive coefficients in the post-COVID model mirrored those in the pre-COVID model, with medical and STEM fields continuing to dominate, and no substantial changes observed in location-based patterns.

However, notable shifts emerged in the negative coefficients, particularly concerning location. While some cities retained their significance from the pre-COVID model, several new cities became significantly negative after COVID-19 compared to Artesia, with stronger coefficients than those seen in disadvantaged academic fields. This shift suggests that, in the post-pandemic period, certain cities, possibly those facing greater economic or social challenges, exerted a more substantial negative impact on salary outcomes than specific academic disciplines.

Overall, the pandemic did not significantly alter societal employment trends, with the exception of some impacts on specific locations. Positive salary associations were consistently observed in healthcare, medical fields, engineering, and STEM disciplines, all of which demonstrated higher earnings compared to Accounting programs. Conversely, negative associations persisted in creative arts, humanities, and niche fields like outdoor education, which yielded lower earnings than Accounting. Additionally, graduates from private (for-profit) and public institutions earned less than those from foreign institutions, with the salary gap being larger for graduates of private (for-profit) institutions.

Post-COVID, economically disadvantaged cities such as Miami Lakes, Pablo, and Alhambra exhibited stronger negative impacts on salaries compared to Artesia, highlighting a shift in emphasis from academic fields to location-based disparities. Prestigious locations, including Stanford and Los Altos Hills, continued to correlate with higher earnings due to the presence of top-tier institutions and industry hubs. Smaller towns, such as Grundy and Yuma, characterized by limited academic and professional opportunities, consistently exhibited lower salary outcomes in both pre- and post-COVID periods.

VII. CONCLUSION

This study provides a comprehensive analysis of the impact of the COVID-19 pandemic on the salaries of US college graduates. We leveraged several machine learning techniques such as K-means clustering, Synthetic Control & Regression Analysis to see if there was a change in salaries and if certain groups were impacted by the pandemic. The results of our study show that although there was some change in salaries for certain sectors & groups during the pandemic, the change overall was not significant, with many groups adapting to structural changes in the job market.

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